

ST. JOSEPH UNIVERSITY IN TANZANIA

BASIC TECHNICIAN CERTIFICATE IN ENGINEERING PROGRAMMES (NTA 4)

SEM: I

CST04101 — BASIC CONCEPTS OF COMPUTERS

Centre Code: **SJCET – 11**

CODE

UNIVERSITY EXAMINATIONS

Reg. No.:

FEBRUARY 2025**Time: 2:30 Hours****Max. Marks: 100****OFFICIAL 4LAZIE DOCUMENT**

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INSTRUCTIONS:

Answer **all questions** in Part A. Answer **any five (5)** questions from Part B. Answer **any five (5)** questions from Part C.

PART – A (10 × 2 = 20 MARKS)*(Answer all questions)*

1. Convert 52 into an octal number.
2. State any four characteristics of a computer.
3. What is an operating system?
4. What are input-output devices in the computer?
5. What is a batch operating system?
6. How are computer systems classified?
7. What are volatile and non-volatile memories?
8. Write the binary and octal equivalent of hexadecimal number **7BD**?
9. What are the basic operations of computers?
10. List out any two (2) kinds of operating system.

PART – B (5 × 4 = 20 MARKS)

(Answer any five)

11. How many types of software are there? Explain them.
12. Differentiate between primary memory and secondary memory.
13. Explain the function and features of BIOS.
14. What is a CPU? Explain the components of the CPU.
15. Convert 278_{10} into a binary number.
16. Write a short note on functions and types of operating system.
17. Explain the difference between impact and non-impact printers.
18. Write any four (4) characteristics of the CPU.
19. What is the difference between magnetic disk and magnetic tape?
20. Differentiate between high level, machine and assembly level language.

PART – C ($5 \times 12 = 60$ MARKS)

(Answer any five)

21. Explain the simple model computer.
22. Explain the following internal and external commands in DOS:
 - (i) DIR
 - (ii) DATE
 - (iii) PATH
 - (iv) CHKDSK
23. Explain the different types of printers.
24. An alarm clock is controlled by a microprocessor. It uses the 24-hour clock. The hour is represented by an 8-bit register, A, and the number of minutes is represented by another 8-bit register, B.
 - (a) Identify what time is represented by the following two 8-bit registers:
 $A = 00010010$ and $B = 00110101$
 - (b) An alarm has been set for 07:30. Two 8-bit registers, C and D, are used to represent the hours and minutes of the alarm time. Show how 07:30 would be represented by these two registers.
25. Explain the classification of the computer by data processed.
26. Explain the visual display unit.
27. Explain primary storage devices.
28. Explain the types of various computers.
29. Convert the following Numbers:
 - (a) $(1010)_2 = (?)_{10}$
 - (b) $(11000011)_2 = (?)_{10}$

(c) $(62)_8 = (?)_{10}$

(d) $(2085)_8 = (?)_{10}$

(e) $(1CE)_{16} = (?)_{10}$

(f) $(2AF)_{16} = (?)_{10}$

30. (a) Explain why modern computers contain both Dynamic RAM (DRAM) and Static RAM (SRAM).
- (b) Computer A has 32 32-bit registers, while Computer B has 16 64-bit registers. Explore two advantages that Computer A possesses over Computer B.

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